Code

Exploring NASA Potentially Hazardous Asteroid Classifications

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Introduction

Near-Earth Objects (NEOs) are asteroids and comets whose orbits bring them near Earth's orbit. NASA classifies some near-Earth asteroids as Potentially Hazardous Asteroids (PHAs), meaning their size and orbital path make them worth continued monitoring. The label sounds alarming, but it does not mean an asteroid is expected to hit Earth. It means the object is large enough and close enough to deserve attention.

This project explores which physical, orbital, and close-approach measurements separate asteroids NASA labels as hazardous from those it does not. We use data wrangling and visualization as the main tools, then use one classification tree as a simple check on which variables are most useful for reproducing NASA's label.

A key skepticism point is that NASA's PHA definition is partly based on absolute magnitude, which is related to asteroid size. We therefore compare a full feature set to a skeptical feature set that removes size and brightness. This gives the report a clear roadmap: describe the data, explore the main variable patterns, test whether the same patterns appear in a simple model, and explain the limits of the findings.

Data

The primary dataset is `neo_full_dataset.xlsx` , a local object-level dataset with one row per asteroid. We use this file instead of `neo.csv` because `neo.csv` is event-like: it contains repeated asteroid IDs from multiple close approaches. Using repeated object rows without grouped splitting could leak the same asteroid into both training and test sets.

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```
library(readxl)
library(dplyr)
library(tidyr)
library(ggplot2)
library(rpart)
library(knitr)
```

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```
data_path <- "neo_full_dataset.xlsx"
if (!file.exists(data_path)) {
  data_path <- "ds-2020-final/neo_full_dataset.xlsx"
}

neo_raw <- read_excel(data_path)

tibble(
  rows = nrow(neo_raw),
  columns = ncol(neo_raw)
) %>%
  kable(caption = "Raw dataset dimensions")
```

Raw dataset dimensions

rows	columns
42919	11

Data Dictionary

Data dictionary for variables used in the report

variable	source	unit	role	notes
neo_reference_id	NASA NeoWs/SBDB extraction	none	Identifier	Used only for duplicate checks, not modeling.
name	NASA NeoWs/SBDB extraction	none	Identifier	Object name; not used for modeling.
absolute_magnitude_h	NASA NeoWs/SBDB extraction	magnitude	Predictor	Brightness measure related to asteroid size; definition-adjacent.
est_diam_min_km	NASA NeoWs/SBDB extraction	kilometers	Raw input	Lower estimated diameter bound; replaced by average diameter.
est_diam_max_km	NASA NeoWs/SBDB extraction	kilometers	Raw input	Upper estimated diameter bound; replaced by average diameter.
diam_avg_km	Engineered	kilometers	Predictor	Mean of minimum and maximum estimated diameter.
miss_distance_km	NASA NeoWs/SBDB extraction	kilometers	Predictor	Close-approach miss distance; related to closeness but not the same as MOID.

variable	source	unit	role	notes
relative_velocity_km_s	NASA NeoWs/SBDB extraction	km/s	Predictor	Relative velocity during close approach.
orbital_eccentricity	NASA SBDB extraction	none	Predictor	Orbital shape.
orbital_inclination	NASA SBDB extraction	degrees	Predictor	Tilt of orbit relative to reference plane.
orbital_semi_major_axis	NASA SBDB extraction	AU	Predictor	Average orbital distance scale.
is_potentially_hazardous_asteroid	NASA NeoWs/SBDB extraction	logical	Target	Converted to the factor variable hazardous.

Missingness and Duplicates

Missing values by variable

variable	missing_n	missing_pct
miss_distance_km	1762	4.1%
relative_velocity_km_s	1762	4.1%
absolute_magnitude_h	13	0.0%
est_diam_min_km	13	0.0%
est_diam_max_km	13	0.0%
name	4	0.0%
orbital_eccentricity	4	0.0%
orbital_inclination	4	0.0%
orbital_semi_major_axis	4	0.0%
neo_reference_id	0	0.0%
is_potentially_hazardous_asteroid	0	0.0%

Duplicate check for the object-level dataset

rows	unique_neo_reference_ids	duplicate_neo_reference_ids
42919	42919	0

Why the event-like CSV was not used as the main modeling table

dataset	rows	unique_ids	interpretation
neo.csv	90836	27423	Event-like table with repeated asteroid IDs

Cleaning and Feature Engineering

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```
model_vars_full <- c(
  "absolute_magnitude_h",
  "diam_avg_km",
  "miss_distance_km",
  "relative_velocity_km_s",
  "orbital_eccentricity",
  "orbital_inclination",
  "orbital_semi_major_axis"
)

model_vars_skeptical <- c(
  "miss_distance_km",
  "relative_velocity_km_s",
  "orbital_eccentricity",
  "orbital_inclination",
  "orbital_semi_major_axis"
)

neo_clean <- neo_raw %>%
  mutate(
    diam_avg_km = (est_diam_min_km + est_diam_max_km) / 2,
    hazardous = factor(
      is_potentially_hazardous_asteroid,
      levels = c(FALSE, TRUE),
      labels = c("Not Hazardous", "Hazardous")
    )
  ) %>%
  select(
    neo_reference_id,
    hazardous,
    all_of(model_vars_full)
  ) %>%
  drop_na(all_of(model_vars_full), hazardous)

tibble(
  raw_rows = nrow(neo_raw),
  cleaned_rows = nrow(neo_clean),
  rows_removed = nrow(neo_raw) - nrow(neo_clean),
  duplicate_ids_after_cleaning = sum(duplicated(neo_clean$neo_reference_id))
) %>%
  mutate(
    rows_removed_pct = format_percent(rows_removed / raw_rows)
  ) %>%
  select(raw_rows, cleaned_rows, rows_removed, rows_removed_pct, duplicate_ids_after_cleaning) %
>%
  kable(caption = "Rows retained after cleaning and feature engineering")
```

Rows retained after cleaning and feature engineering

raw_rows	cleaned_rows	rows_removed	rows_removed_pct	duplicate_ids_after_cleaning

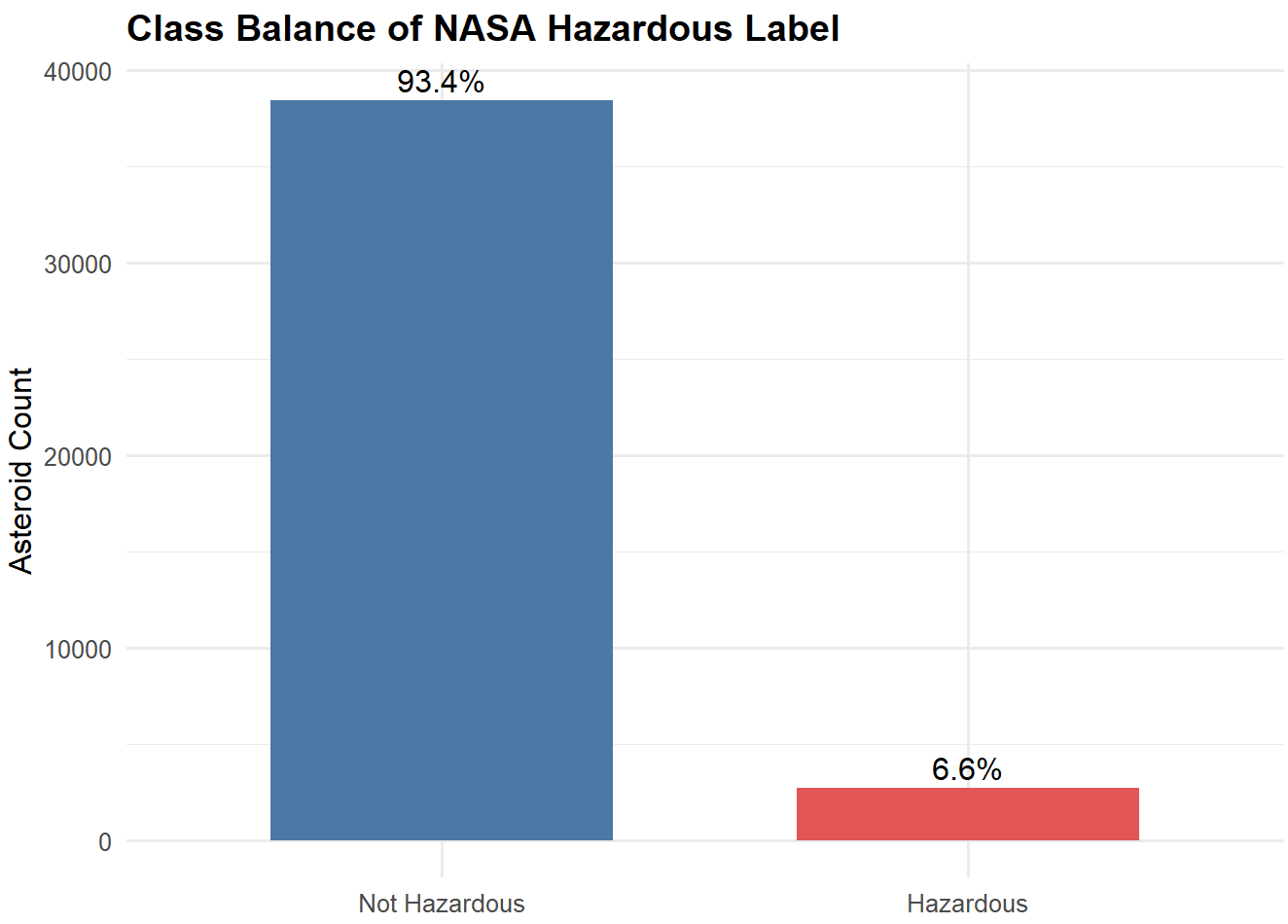
raw_rows	cleaned_rows	rows_removed	rows_removed_pct	duplicate_ids_after_cleaning
42919	41157	1762	4.1%	0

Exploratory Analysis

Class Balance

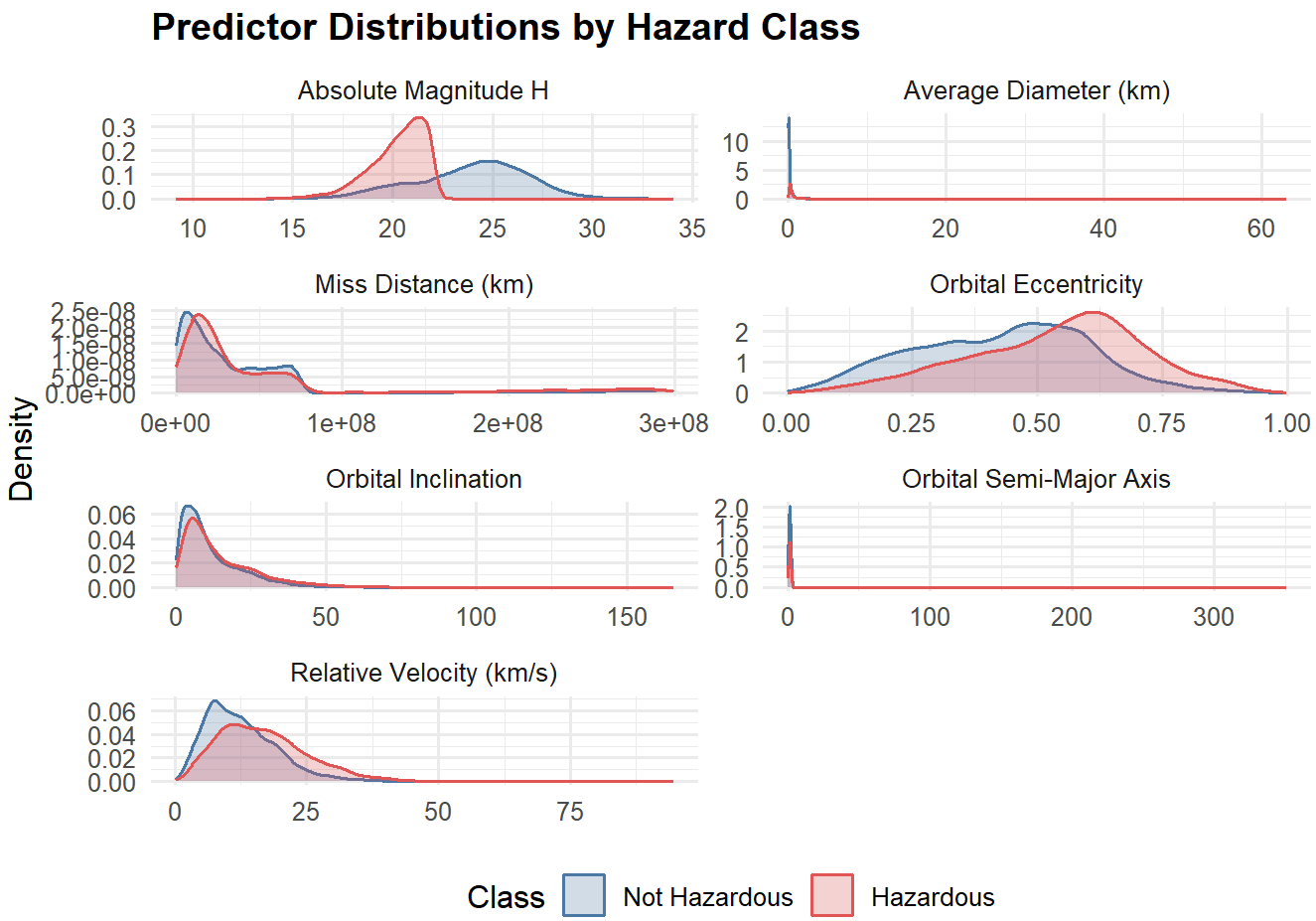
Class balance in the cleaned asteroid dataset

hazardous	n	prop
Not Hazardous	38434	93.4%
Hazardous	2723	6.6%



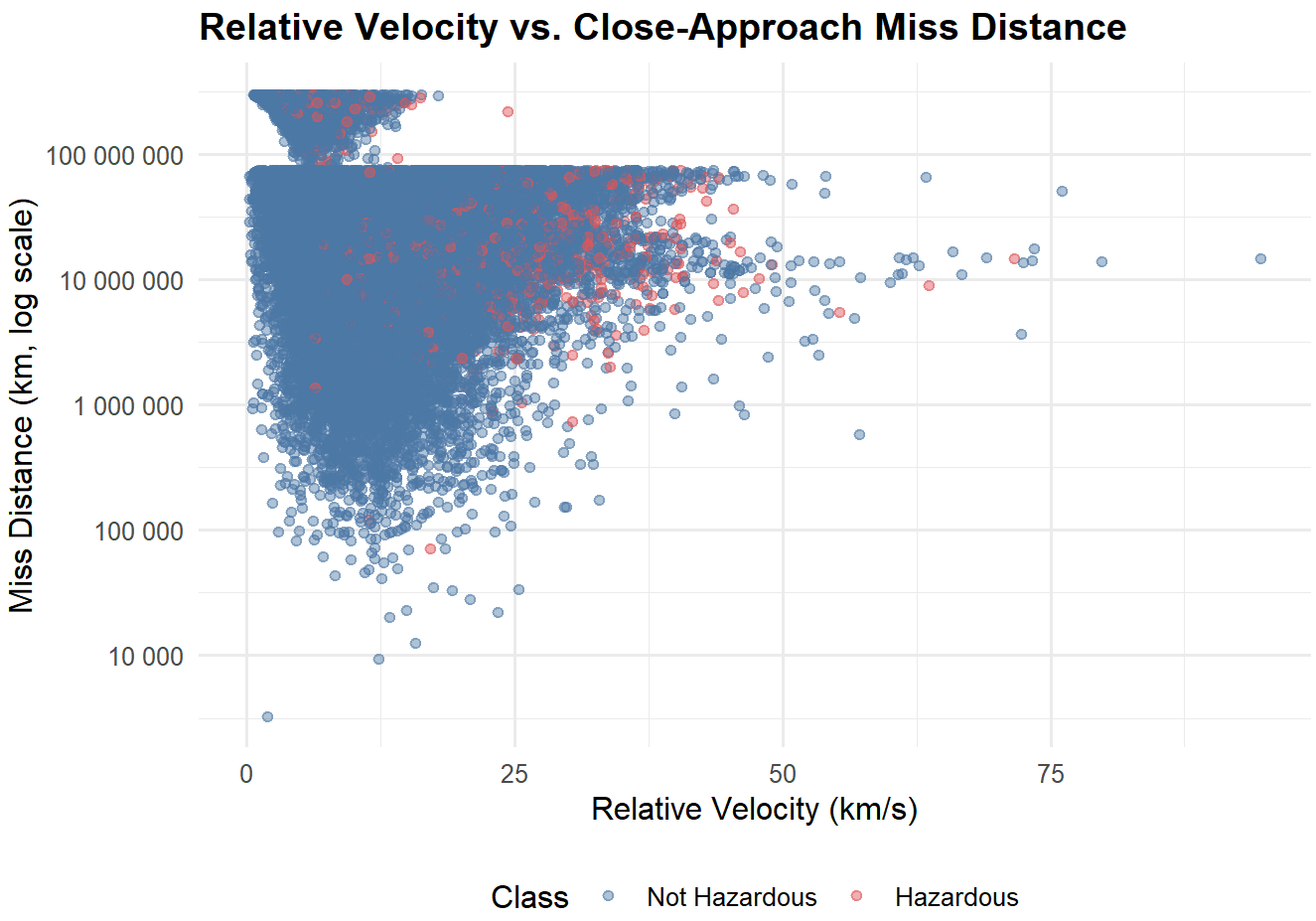
The hazardous class is much smaller than the non-hazardous class, so accuracy alone is not a reliable evaluation metric. A model can look accurate while missing many hazardous objects.

Feature Distributions by Hazard Class



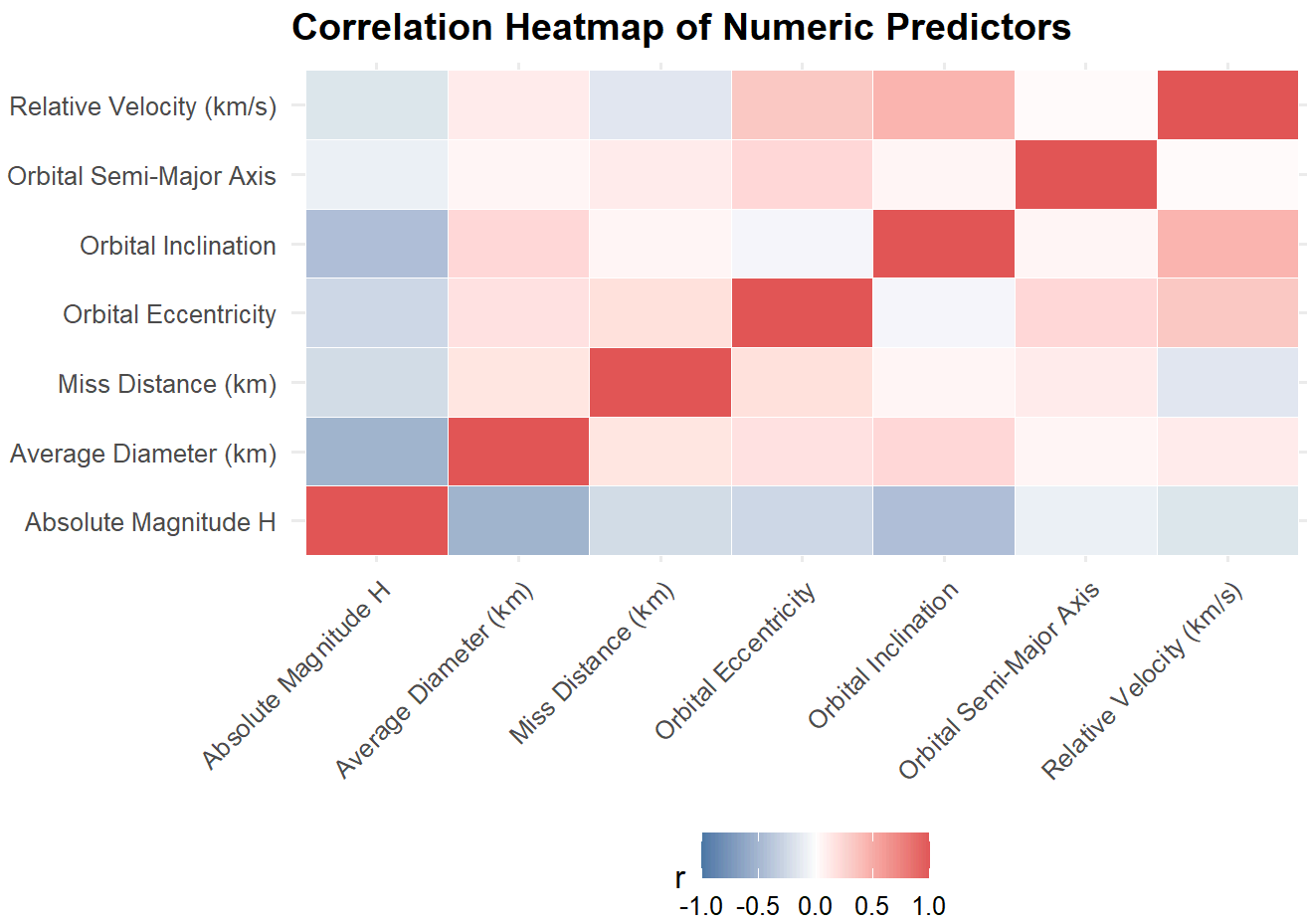
The clearest separation appears in size-related variables: hazardous asteroids tend to have lower absolute magnitude values and larger estimated diameters. The orbital and close-approach variables show more overlap, which suggests that NASA's label is not explained by one simple measurement in this dataset.

Relative Velocity vs. Miss Distance



This scatter plot shows why the problem is not as simple as “closer asteroids are always hazardous.” Hazardous and non-hazardous asteroids overlap across much of the velocity and miss-distance range.

Correlation Heatmap



Average diameter and absolute magnitude are strongly related, which matches the science: brighter asteroids are usually larger. This correlation is also part of why we avoid treating size and brightness as independent evidence in the skeptical model below.

Analysis Decisions and Trials

During earlier drafts, we tried a more model-heavy version of the analysis with logistic regression, classification trees, ROC curves, and precision-recall curves. Those tools did not change the main finding: size and brightness carry much of the signal, while the remaining orbital and close-approach variables are weaker by themselves. We therefore kept the final report focused on visual exploration and one interpretable classification tree. We also chose the object-level Excel file over the event-like CSV because repeated asteroid IDs in the CSV could make model evaluation look better than it really is.

Modeling

We use a classification tree because it is easier to explain than a more mathematical model: the tree sorts asteroids by splitting on variables that best separate hazardous from non-hazardous labels. The tree is fit twice:

- **Full feature set:** orbital, close-approach, size, and brightness predictors.
- **Skeptical feature set:** orbital and close-approach predictors only.

The skeptical feature set tests whether the model can still reproduce NASA's label when we remove the variables most closely connected to the official PHA definition.

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```
set.seed(2026)

neo_model <- neo_clean %>%
  mutate(row_id = row_number())

train_ids <- neo_model %>%
  group_by(hazardous) %>%
  slice_sample(prop = 0.80) %>%
  ungroup() %>%
  pull(row_id)

train_data <- neo_model %>%
  filter(row_id %in% train_ids) %>%
  select(-row_id, -neo_reference_id)

test_data <- neo_model %>%
  filter(!row_id %in% train_ids) %>%
  select(-row_id, -neo_reference_id)

tibble(
  split = c("Train", "Test"),
  rows = c(nrow(train_data), nrow(test_data)),
  hazardous_rate = c(
    mean(train_data$hazardous == "Hazardous"),
    mean(test_data$hazardous == "Hazardous")
  )
) %>%
  mutate(hazardous_rate = format_percent(hazardous_rate)) %>%
  kable(caption = "Stratified train/test split")
```

Stratified train/test split

split	rows	hazardous_rate
Train	32925	6.6%
Test	8232	6.6%

Evaluation Functions

We use four confusion-matrix metrics. Accuracy is the percent of all predictions that are correct. Recall is the percent of truly hazardous asteroids the model catches. Specificity is the percent of non-hazardous asteroids the model correctly leaves alone. Precision is the percent of predicted hazardous asteroids that are actually hazardous.

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```
safe_divide <- function(numerator, denominator) {
  ifelse(denominator == 0, NA_real_, numerator / denominator)
}

confusion_metrics <- function(actual, predicted, positive = "Hazardous") {
  actual <- as.character(actual)
  predicted <- as.character(predicted)
  negative <- setdiff(unique(actual), positive)[1]

  tp <- sum(actual == positive & predicted == positive)
  tn <- sum(actual == negative & predicted == negative)
  fp <- sum(actual == negative & predicted == positive)
  fn <- sum(actual == positive & predicted == negative)

  tibble(
    tp = tp,
    tn = tn,
    fp = fp,
    fn = fn,
    accuracy = safe_divide(tp + tn, tp + tn + fp + fn),
    recall = safe_divide(tp, tp + fn),
    specificity = safe_divide(tn, tn + fp),
    precision = safe_divide(tp, tp + fp)
  )
}
```

Fit Classification Trees

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```
full_formula <- as.formula(
  paste("hazardous ~", paste(model_vars_full, collapse = " + "))
)

skeptical_formula <- as.formula(
  paste("hazardous ~", paste(model_vars_skeptical, collapse = " + "))
)

tree_full <- rpart(
  full_formula,
  data = train_data,
  method = "class",
  control = rpart.control(cp = 0.001, minbucket = 25)
)

tree_skeptical <- rpart(
  skeptical_formula,
  data = train_data,
  method = "class",
  control = rpart.control(cp = 0.001, minbucket = 25)
)

tree_full_predicted <- predict(tree_full, newdata = test_data, type = "class")
tree_skeptical_predicted <- predict(tree_skeptical, newdata = test_data, type = "class")
```

Results

Tree Comparison

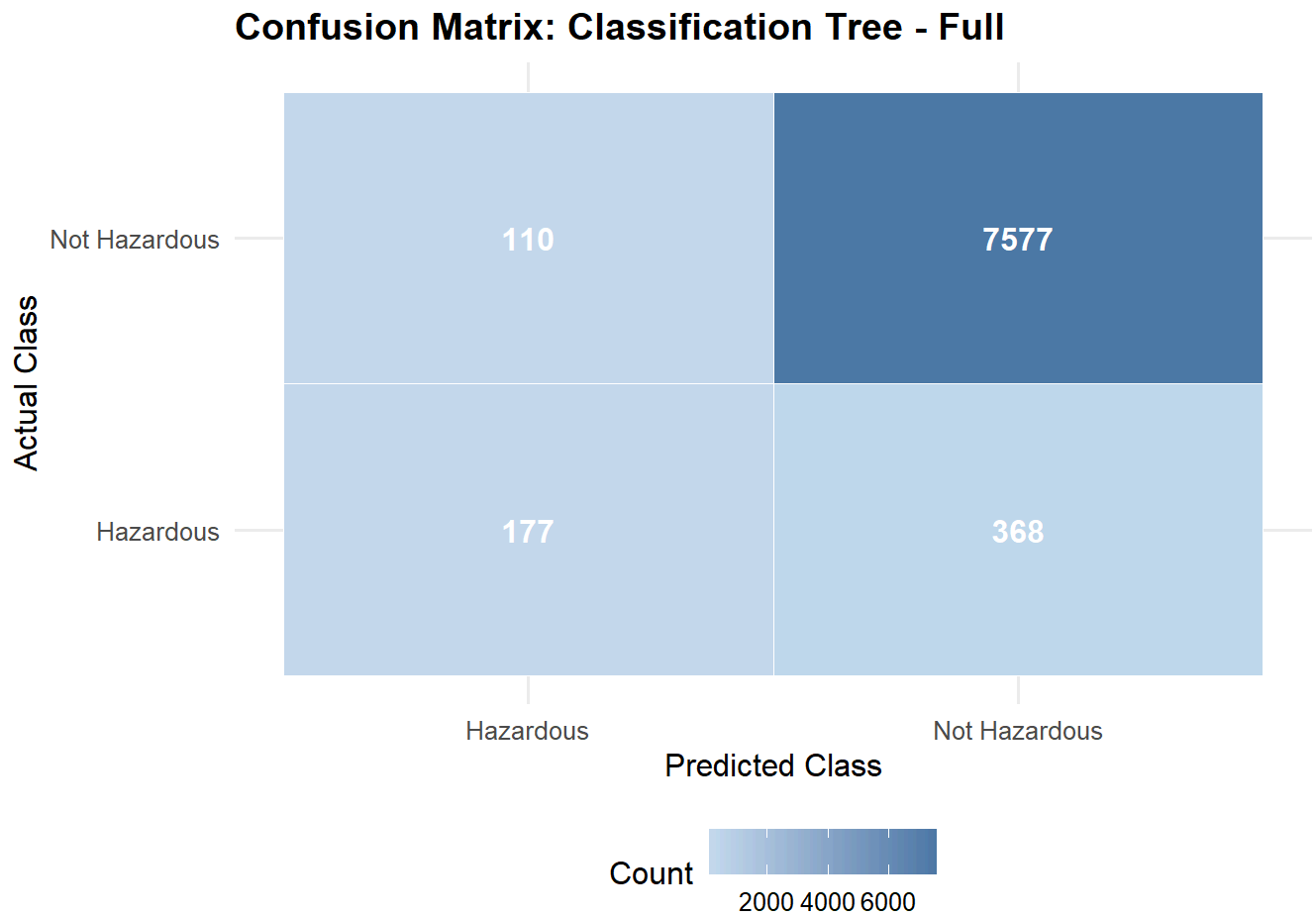
Classification tree performance on the test data

Feature Set	Accuracy	Recall	Specificity	Precision	True Positives	True Negatives	False Positives	False Negatives	Note
Full	94.2%	32.5%	98.6%	61.7%	177	7577	110	368	**Highest recall**
Skeptical	93.4%	0.0%	100.0%	NA	0	7687	0	545	

The full tree performs better at identifying hazardous asteroids. The skeptical tree has similar overall accuracy, but it catches none of the hazardous objects in the test set. This is an important result: without size and brightness, the tree mostly learns to predict the majority “Not Hazardous” class.

Confusion Matrix Heatmap

The heatmap below uses the full classification tree, which has the best recall for hazardous asteroids.

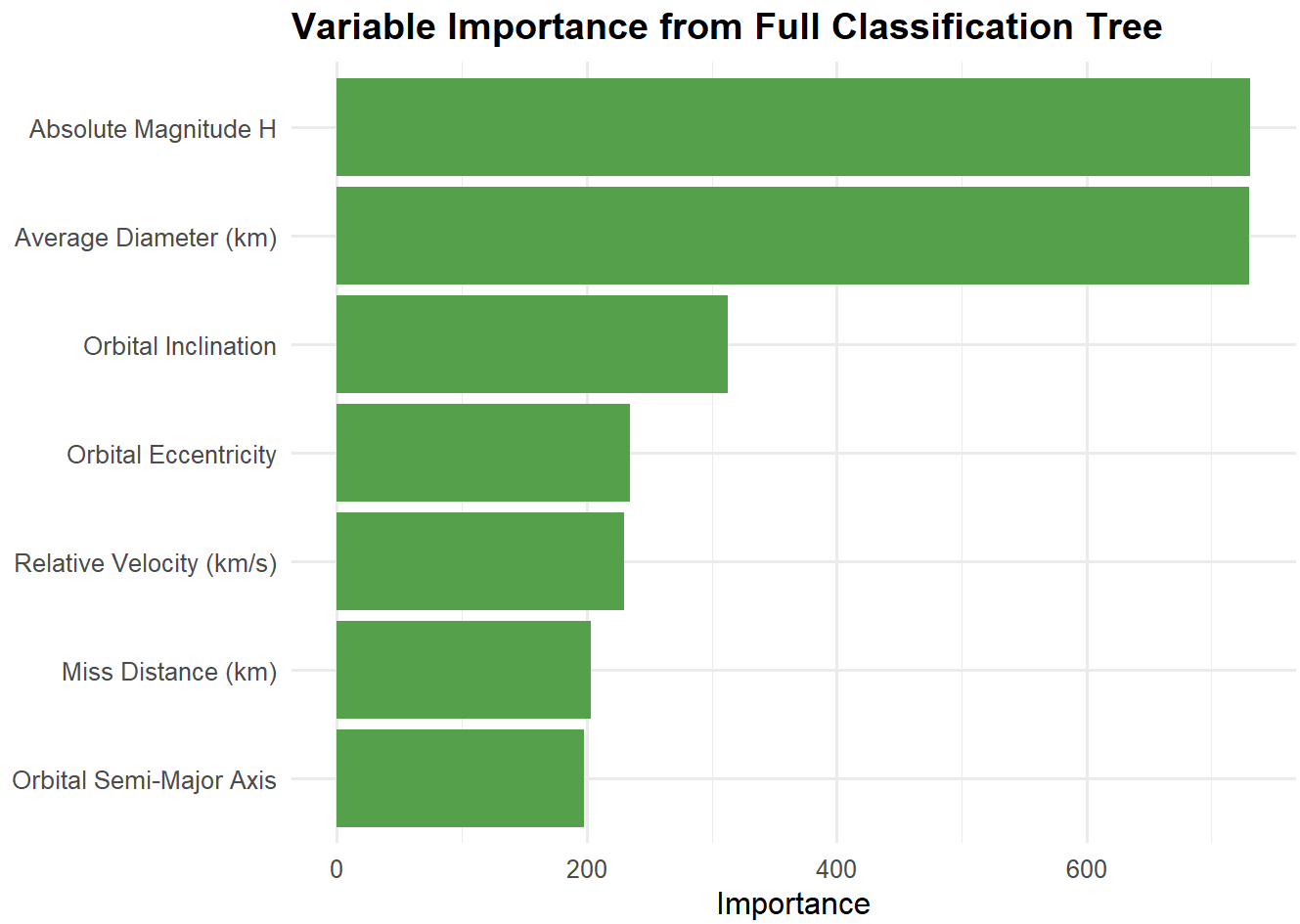


The full tree correctly identifies some hazardous asteroids, but it still misses many of them. This keeps the result honest: the model helps show which variables matter, but it would not be acceptable as a real-world asteroid warning system.

Classification Tree Variable Importance

Most important variables in the full classification tree

variable	importance
Absolute Magnitude H	731.0
Average Diameter (km)	730.4
Orbital Inclination	313.2
Orbital Eccentricity	234.7
Relative Velocity (km/s)	229.7
Miss Distance (km)	203.6
Orbital Semi-Major Axis	197.4



The model's most important variables reinforce the visual findings: size and brightness are central to the hazardous label, while the other orbital variables provide additional but weaker separation.

Skepticism and Limitations

The strongest limitation is that the hazardous label is not an independent scientific outcome like a future impact. It is NASA's classification, and that classification is partly based on variables related to asteroid size and brightness. NASA/JPL CNEOS defines PHAs using Earth MOID of 0.05 AU or less and absolute magnitude H of 22.0 or brighter (CNEOS NEO Groups (https://cneos.jpl.nasa.gov/about/neo_groups.html)). Absolute magnitude is therefore definition-adjacent, and average diameter is closely related to the same idea.

The dataset also does not include Earth MOID, the official orbital-distance criterion used in the PHA definition. We use `miss_distance_km`, but close-approach miss distance is not the same as MOID. This means the model approximates NASA's classification using the available object-level variables rather than exactly applying NASA's rule.

Other limitations:

- The hazardous class is imbalanced, so accuracy can be misleading.
- Rows with missing modeling fields were removed rather than imputed.
- The close-approach variables appear to summarize one approach field, not the full future distribution of possible close approaches.
- The models predict NASA's label; they do not estimate causal asteroid danger or actual Earth impact probability.

Conclusion

The strongest pattern in the data is that NASA's hazardous label is closely tied to asteroid size and brightness. Hazardous asteroids are rare in the cleaned dataset, making up about 6.6% of observations, and the full classification tree catches more hazardous objects than the skeptical tree. When size and brightness are removed, the model mostly defaults to predicting the majority non-hazardous class.

The main takeaway is therefore not that we built a danger-prediction system. Instead, we showed that NASA's PHA label can be partly reproduced from available asteroid measurements, while also showing why that result must be interpreted carefully. The model is strongest when it includes variables that are close to NASA's own definition.

Future work should add Earth MOID, the official orbital-distance part of the PHA definition, and should compare multiple close approaches for the same asteroid rather than relying on one summarized approach field. It would also be useful to separate asteroids by size group or orbit type to see whether the same patterns hold within more specific asteroid categories.